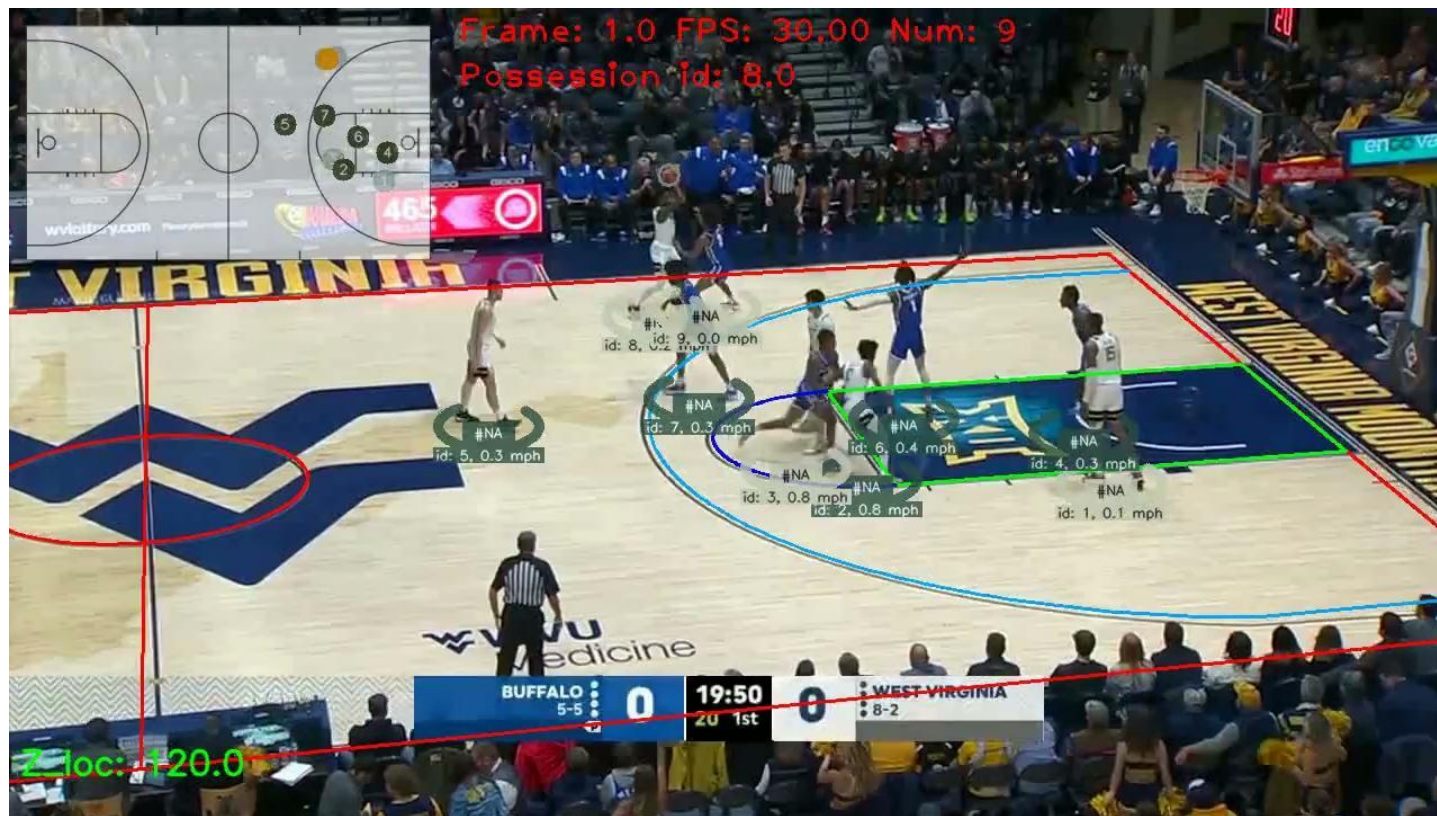


Extracting Tracking Data from Single View Broadcast Video

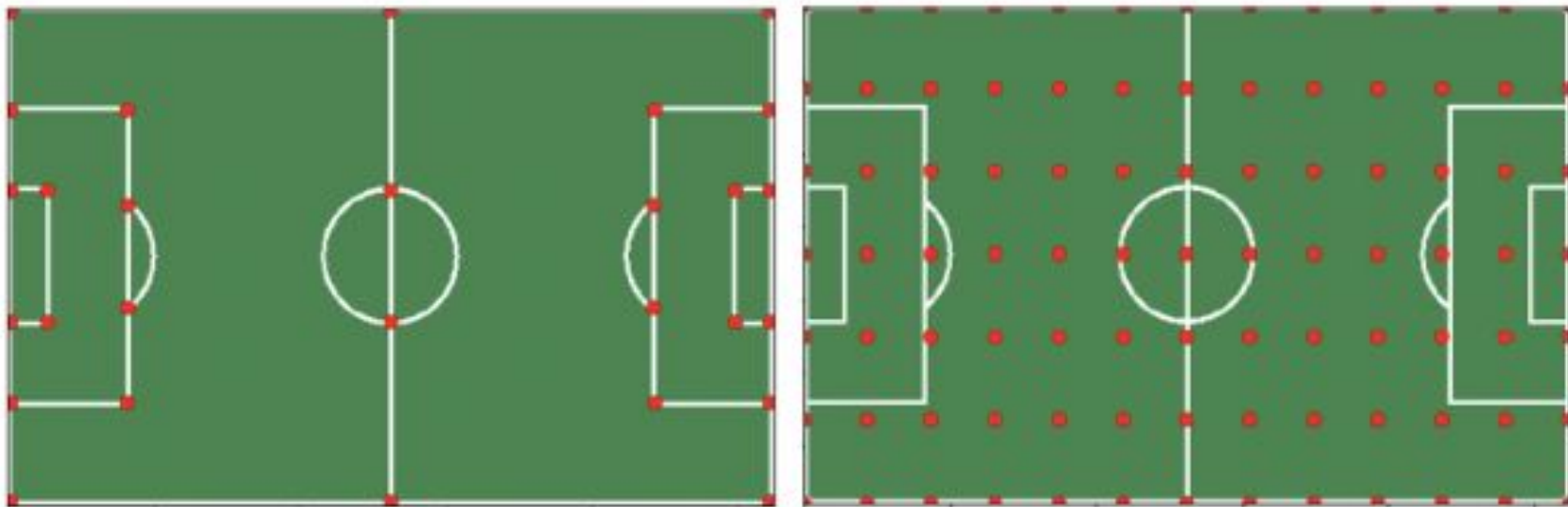
Simon Minami

Project Overview



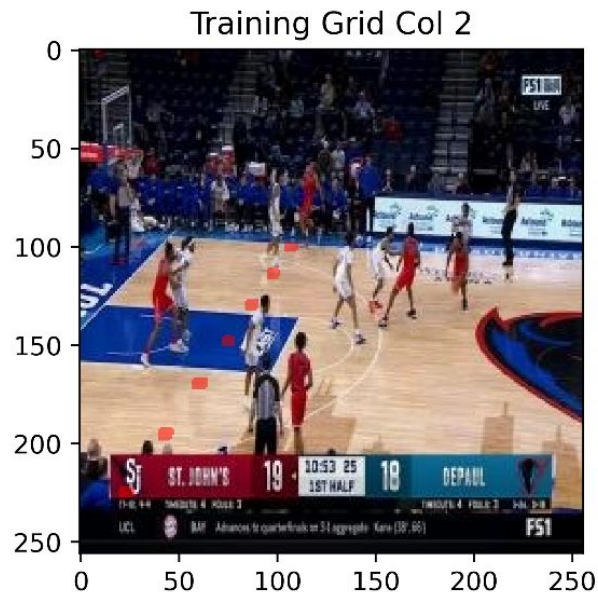
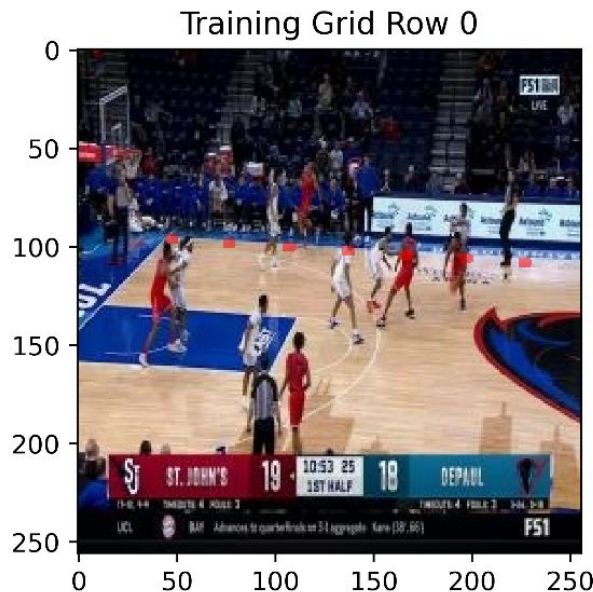
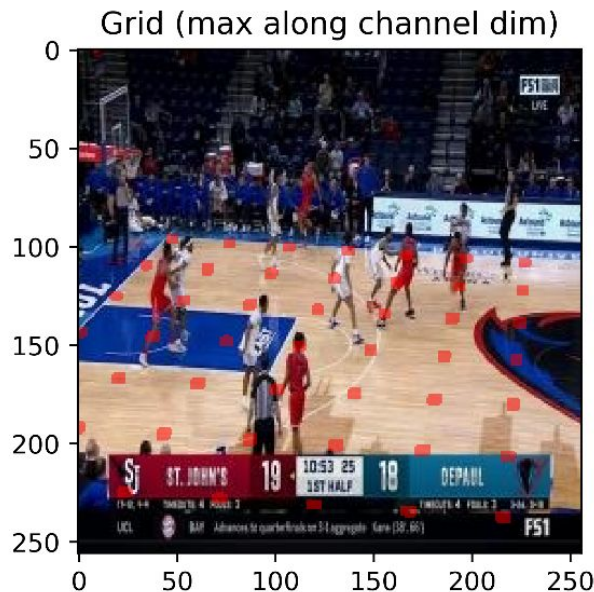
Homography

Which Keypoints to Predict?



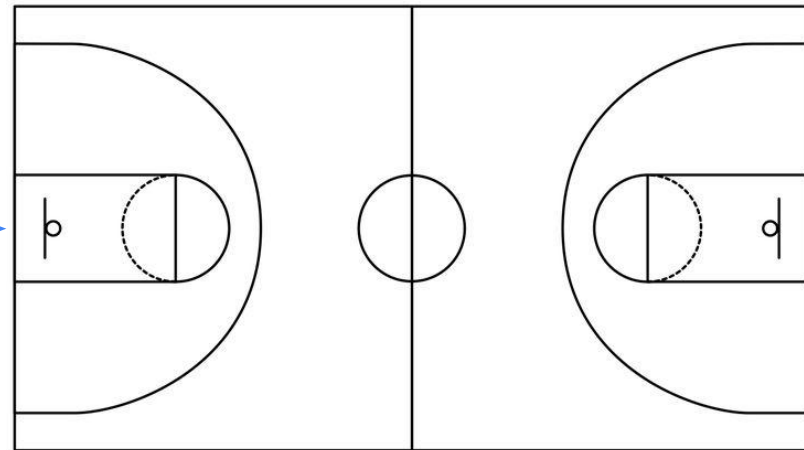
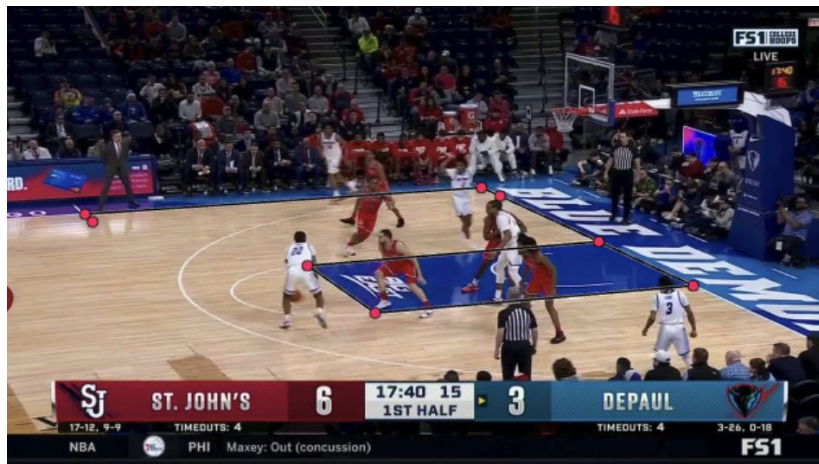
How do we actually represent the keypoints?

- 22, H, W heatmaps



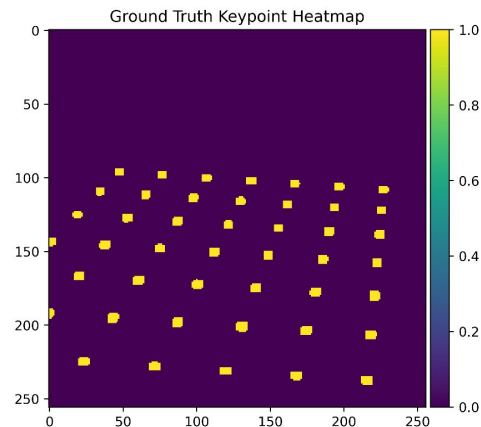
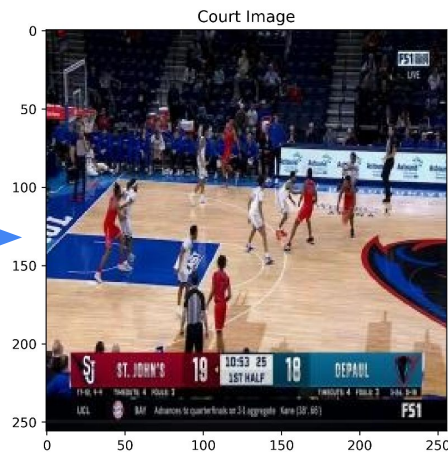
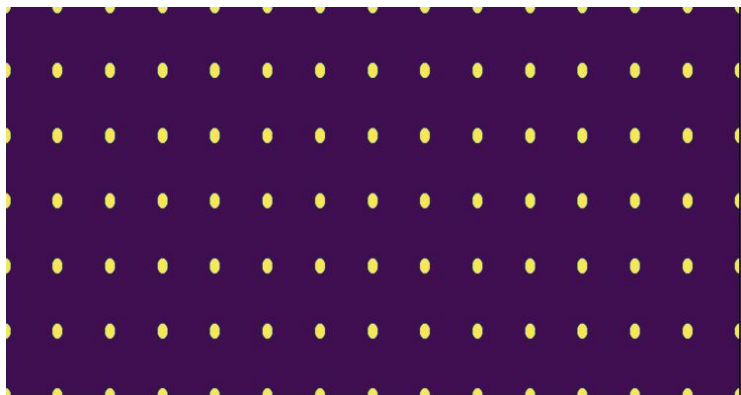
Annotation Process

- Annotate known points, calculate ground truth homography [H](#)



How do we generate the Training Data? (keypoint heatmaps)

- Use H^{-1} to project template to 22, H, W keypoint heatmap



Model

- Dense per pixel prediction
- Net encoder–decoder
 - downsampling to a semantic latent space, then upsampling while using skip connections to recover fine detail needed to localize small points.

Losses

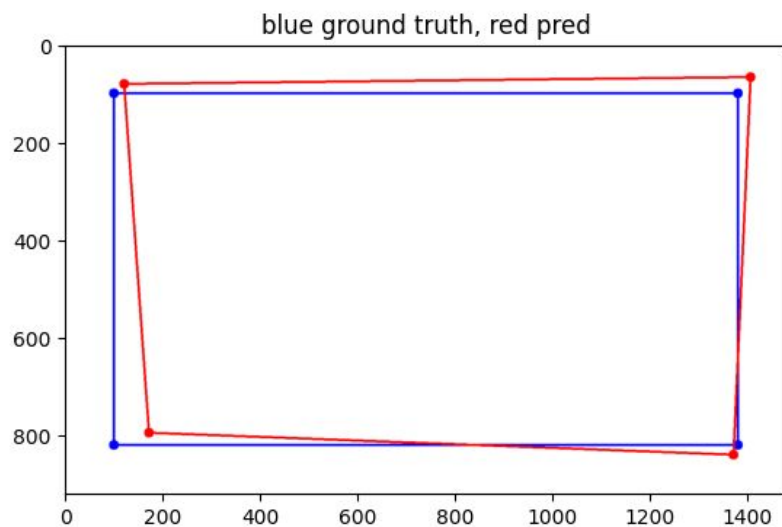
- Pixel-wise cross-entropy on the row and column heatmaps but only at ground truth keypoint locations, to classify each visible keypoint's row and column.
- To discourage spurious detections, we collapsed the heatmaps and added a binary cross-entropy mask loss over keypoint vs background.

Decoding

- Post-processing finds peaks in the predicted heatmaps (local maxima).
- We assign each detected pixel to its most likely row and column (via argmax across row/col channels) to recover the full 7×15 grid mapping.

Evaluation

- IOU, reprojection error



Next Steps

- Our current U-Net is limited to local convolutions; to better model the global grid structure and long-range relationships, we'd explore self-attention in the bottleneck (e.g., TransUNet-style) or other transformer blocks.
- Could also experiment with fusing other cues (e.g., explicit line detection or semantic segmentation of the court) to anchor predictions.